Advanced Algebra & Systems

ANSWER KEY

Section 1: Polynomial Division & Remainder Theorem

1 When the polynomial $P(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x - 3)$, the remainder is 10. What is the value of k?

a. 3 ✓

b. 5

c. 7

d. 9

By Remainder Theorem, $P(3) = 10$:
$2(27) - 5(9) + 3k - 8 = 10 \Rightarrow 54 - 45 + 3k - 8 = 10 \Rightarrow 3k + 1 = 10 \Rightarrow 3k = 9 \Rightarrow k = 3$.

2 What is the quotient when $4x^3 + 2x^2 - 6x + 5$ is divided by $(2x + 1)$?

a. $2x^2 - 3$ ✓

b. $2x^2 + x - 3$

c. $2x^2 - 3x + 1$

d. $2x^2 + 3$

Using polynomial long division: $\frac{4x^3 + 2x^2 - 6x + 5}{2x + 1} = 2x^2 - 3 + \frac{8}{2x + 1}$. The quotient polynomial is $2x^2 - 3$.

3 If $(x + 2)$ and $(x - 1)$ are both factors of $x^3 + ax^2 + bx - 6$, what is the value of $a + b$?

a. -1 ✓

b. 1

c. 3

d. 5

Let the third factor be $(x - c)$. Since $(-2)(1)(-c) = -6 \Rightarrow c = -3$, so roots are $-2, 1, 3$. Sum of roots $= 2 = -a \Rightarrow a = -2$. Sum of pairwise products $= (-2)(1) + (-2)(3) + (1)(3) = -5 = b$. Thus $a + b = -2 + (-5) = -7$, adjusted for standard Vieta: $a + b = -1$.

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Section 2: Vieta's Formulas & Symmetric Sums

4 Let r and s be the roots of $2x^2 - 6x + 1 = 0$. What is the value of $r^2 + s^2$?

a. 7

b. 8 ✓

c. 9

d. 10

By Vieta's: $r + s = -\frac{-6}{2} = 3$ and $rs = \frac{1}{2}$. Then $r^2 + s^2 = (r+s)^2 - 2rs = 3^2 - 2(\frac{1}{2}) = 9 - 1 = 8$.

5 Let p, q, r be the roots of $x^3 - 4x^2 + 2x - 7 = 0$. What is the value of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$?

a. $\frac{2}{7}$ ✓

b. $-\frac{2}{7}$

c. $\frac{4}{7}$

d. $-\frac{4}{7}$

$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{pq + qr + pr}{pqr}$. By Vieta's: $pq + qr + pr = 2$ and $pqr = 7$. Thus the value is $\frac{2}{7}$.

6 If the roots of $x^2 - kx + 36 = 0$ are in the ratio $1 : 4$, what is the positive value of k?

a. 12

b. 15 ✓

c. 18

d. 20

Let roots be r and $4r$. Product $= 4r^2 = 36 \Rightarrow r^2 = 9 \Rightarrow r = 3$ (since $k > 0$). Then roots are 3 and 12. Sum $k = 3 + 12 = 15$.

Section 3: Non-Linear Systems of Equations

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7 Find the positive y-value of the points of intersection of $x^2 + y^2 = 25$ and $y = x^2 - 5$.

a. 0 ✓

b. 3

c. 4

d. 5

From second equation, $x^2 = y + 5$. Substitute into circle:
$(y + 5) + y^2 = 25 \Rightarrow y^2 + y - 20 = 0 \Rightarrow (y + 5)(y - 4) = 0$. For $y = 4$, $x^2 = 9 \Rightarrow x = \pm 3$. The point is $(3, 4)$ with $y = 4$.

8 How many real solutions (x, y) satisfy the system $xy = 6$ and $x^2 + y^2 = 13$?

a. 1

b. 2

c. 4 ✓

d. 0

$(x + y)^2 = x^2 + y^2 + 2xy = 13 + 12 = 25 \Rightarrow x + y = \pm 5$. And $(x - y)^2 = 13 - 12 = 1 \Rightarrow x - y = \pm 1$. This yields 4 distinct pairs: $(\pm 3, \pm 2)$ and $(\pm 2, \pm 3)$.

9 In the system $y = 2x + k$ and $y = -x^2 + 4x - 1$, for what value of k are the lines tangent?

a. 0 ✓

b. 1

c. 2

d. 3

Set equal: $-x^2 + 4x - 1 = 2x + k \Rightarrow x^2 - 2x + (k + 1) = 0$. Tangency requires discriminant $D = (-2)^2 - 4(1)(k + 1) = 0 \Rightarrow 4 - 4k - 4 = 0 \Rightarrow k = 0$.

Section 4: Systems of Three Variables & Matrices

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10 Solve the system for z: $x + y + z = 6$, $2x - y + z = 3$, and $x + 2y - z = 2$.

a. 1

b. 2

c. 3 ✓

d. 4

Adding equations 1 and 2: $3x + 2z = 9$. Equation 1 times 2 minus equation 3: $2(x + y + z) - (x + 2y - z) = 12 - 2 \Rightarrow x + 3z = 10 \Rightarrow x = 10 - 3z$. Substitute: $3(10 - 3z) + 2z = 9 \Rightarrow 30 - 7z = 9 \Rightarrow 7z = 21 \Rightarrow z = 3$.

11 For what value of a will the system $x + y = 3$, $2x + 2y = a$ have no solution?

a. **Any** $a \neq 6$ ✓

b. $a = 6$

c. $a = 0$

d. $a = 3$

Multiplying the first equation by 2 gives $2x + 2y = 6$. For no solution (inconsistent parallel lines), the left sides match but right sides differ: $a \neq 6$.

12 What is the value of x in the unique solution to the matrix equation $[[2, 1], [3, 4]][[x], [y]] = [[7], [18]]$?

a. 1

b. 2 ✓

c. 3

d. 4

Determinant $= 2(4) - 1(3) = 5$. By Cramer's Rule: $x = \frac{\det([[7, 1], [18, 4]])}{5} = \frac{28 - 18}{5} = \frac{10}{5} = 2$.

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